

SMART FACULTY ACCESS & STUDENT ASSISTANT SYSTEM (LECSTU)



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Introduction & Motivation

- University systems are fragmented and manual
- Limited AI-driven automation
- No integrated multilingual voice + chatbot support
- Poor accessibility for students and faculty

Literature Review & Research Gap

Existing Systems & Technologies

- Moodle – Focuses on content delivery and course management
- Google Assistant – Provides speech-based interaction but not designed for academic workflows.
- Rasa – Enables chatbot development, typically evaluated in isolated domains.
- mBART – Supports multilingual translation, mainly tested on general datasets rather than university-specific contexts..

Identified Research Gap

- No unified AI ecosystem integrating ASR, NLP chatbot, and multilingual translation within a university system.
- Limited research on English–Tamil–Sinhala AI integration in academic environments.
- Lack of empirical usability evaluation of AI-assisted university platforms.

Research Objectives

- RO1 - Evaluate multilingual ASR systems
- RO2 - Develop and evaluate NLP chatbot
- RO3 - Compare translation models
- RO4 - Measure usability impact
- RO5 - Develop integrated LECSTU platform



Methodology & Research Design

Research Approach

- Design Science Research (DSR) methodology
- Develop and evaluate an AI-integrated university system

System Development

- Development of LECSTU platform (Faculty & Student modules)
- Integration of:
 - Speech Recognition (ASR).
 - NLP Chatbot
 - Multilingual Translation

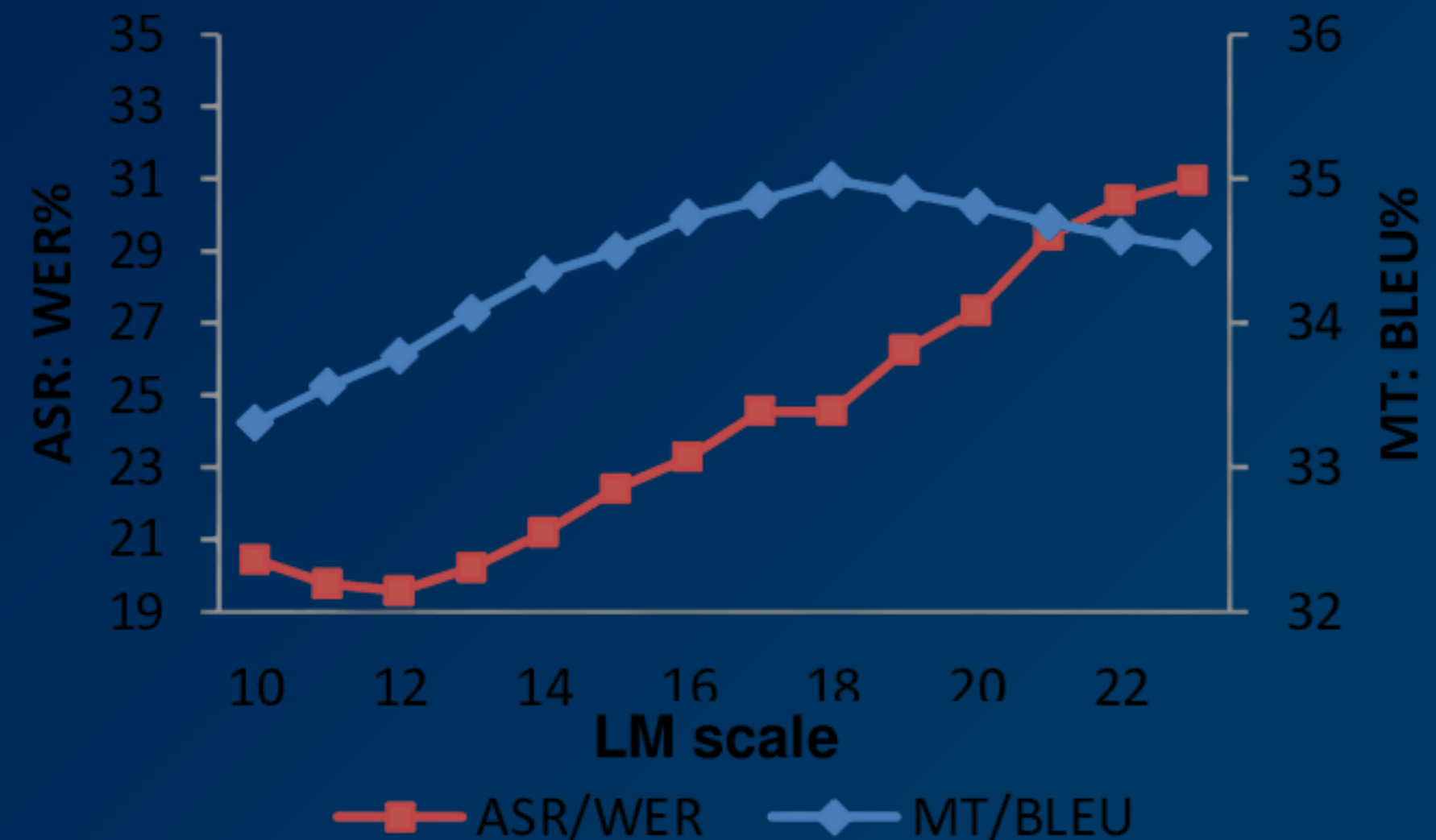


Experimental Evaluation

- ASR evaluated using Word Error Rate (WER)
- Chatbot evaluated using Accuracy and F1-score
- Translation evaluated using BLEU score

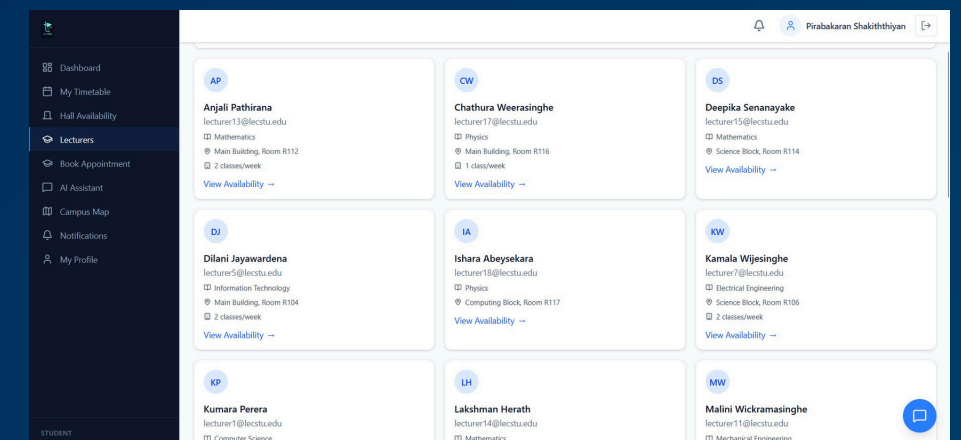
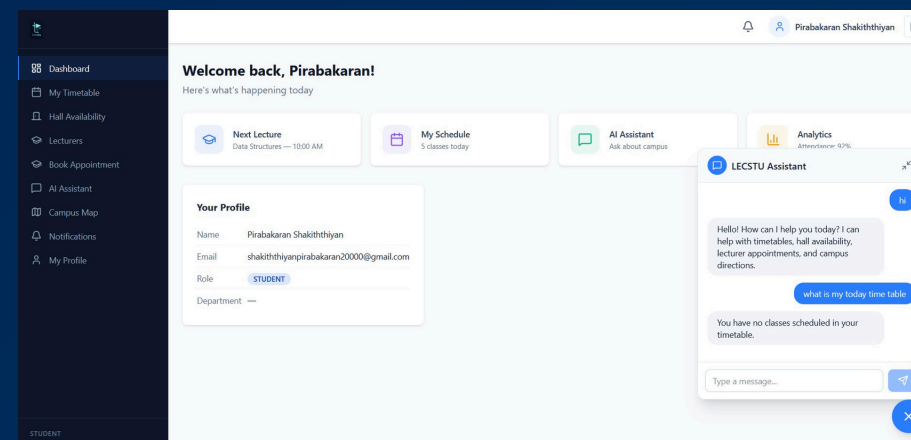
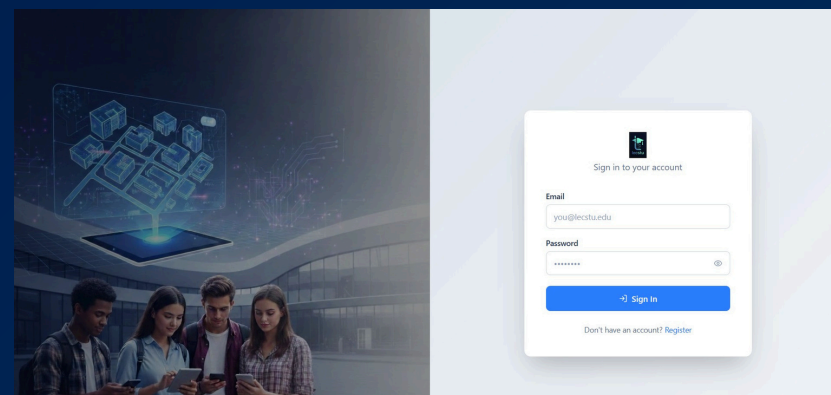
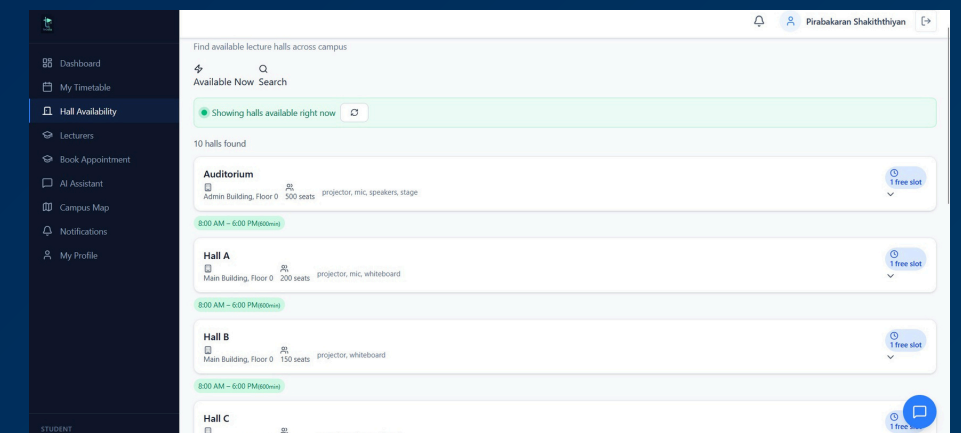
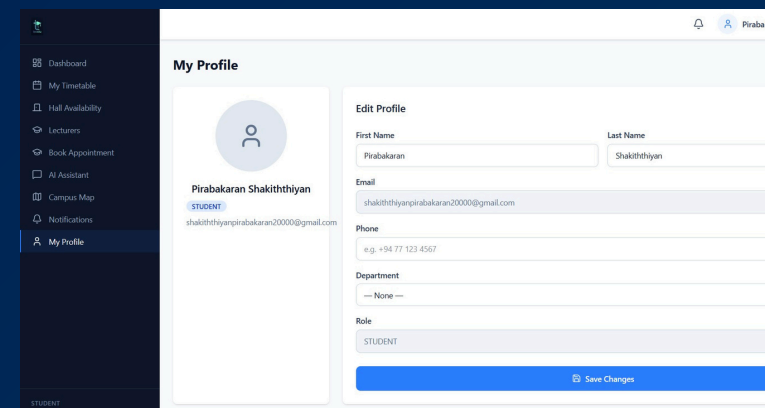
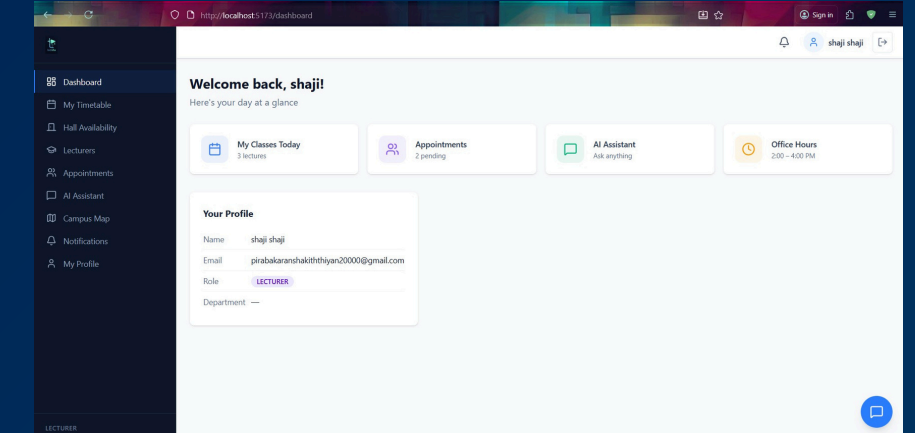
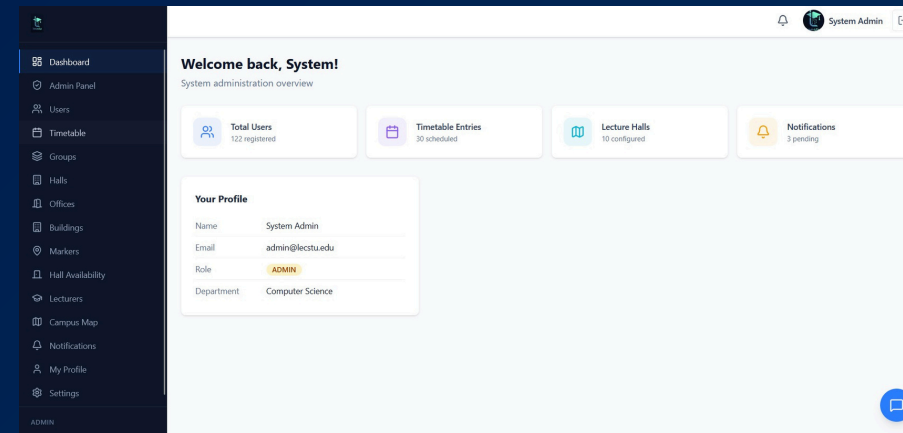
Usability Study

- Within-subject design
- 20+ participants
- System Usability Scale (SUS)



System Development Progress (R05)

- Completed System Modules
- Role-based authentication
- Faculty dashboard
- Student dashboard
- Appointment booking system
- Hall availability module
- Notification system



ASR Evaluation (R01)

- Model Evaluated
 - Whisper
- Evaluation Performed
 - Compared multiple ASR systems
 - Measured Word Error Rate (WER)
 - Measured latency
 - Conducted statistical testing

NLP Chatbot (R02)

- Framework Used
 - Rasa
- Current Progress
 - Intent design completed
 - Training dataset prepared
 - Model trained
 - Evaluation ongoing

Translation System (RO3)

Models

mBART

Baseline cloud API

Current Progress

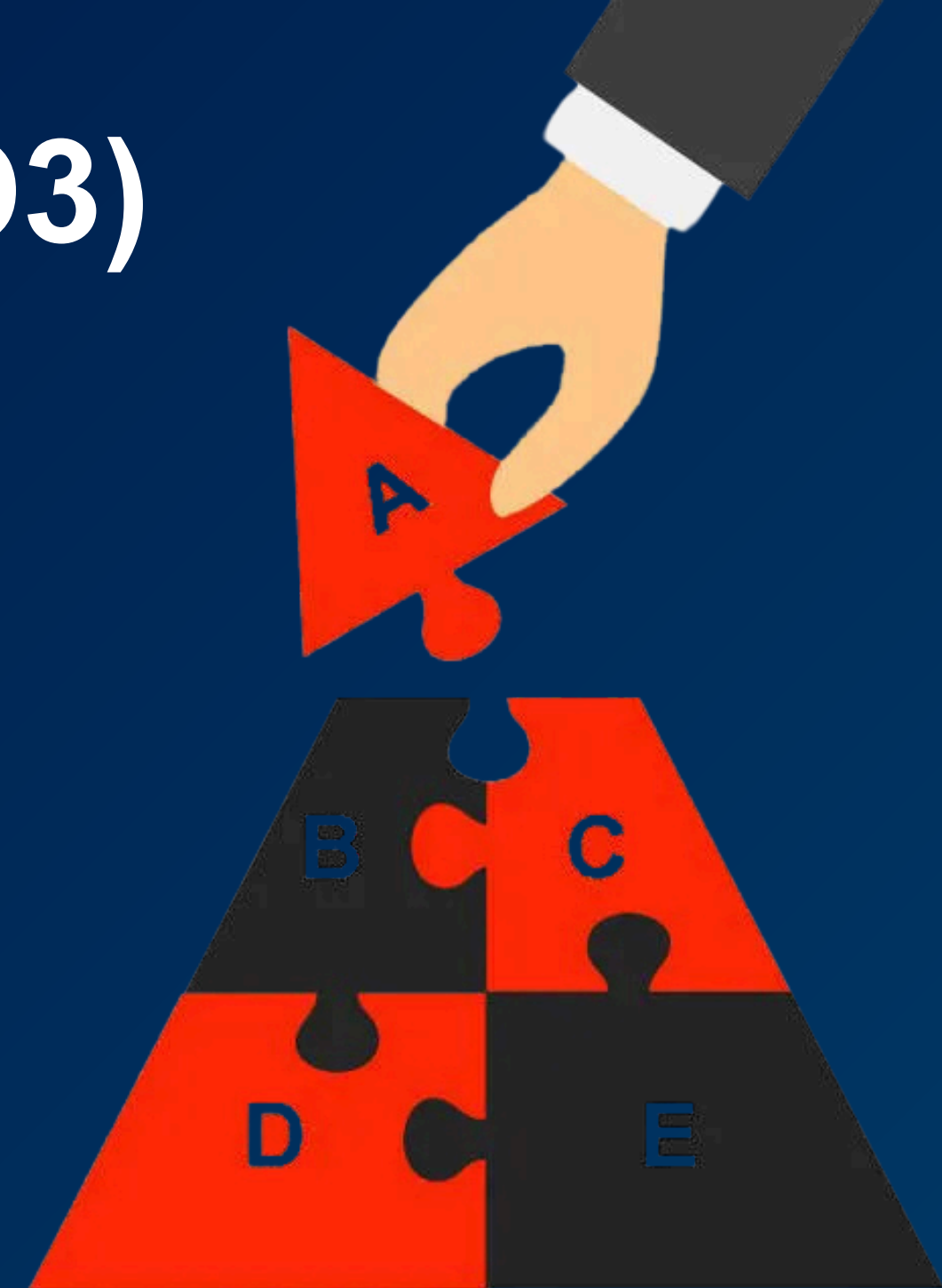
Parallel dataset prepared

BLEU and similarity metrics defined

Performance testing ongoing

Progress vs Timeline

Completed	Ongoing	Remaining
• System development • ASR evaluation	• NLP evaluation • Translation benchmarking	• Usability study • Final statistical validation



Conclusion

The LECSTU system is fully developed, with AI modules successfully integrated. ASR evaluation has been completed, and NLP and translation components are progressing well. The project remains on schedule, and the next steps involve completing AI evaluations, conducting a usability study, and finalizing statistical analysis.



Thank You